UNIT 5: Axial Flow Turbine and CFD for Turbomachinery

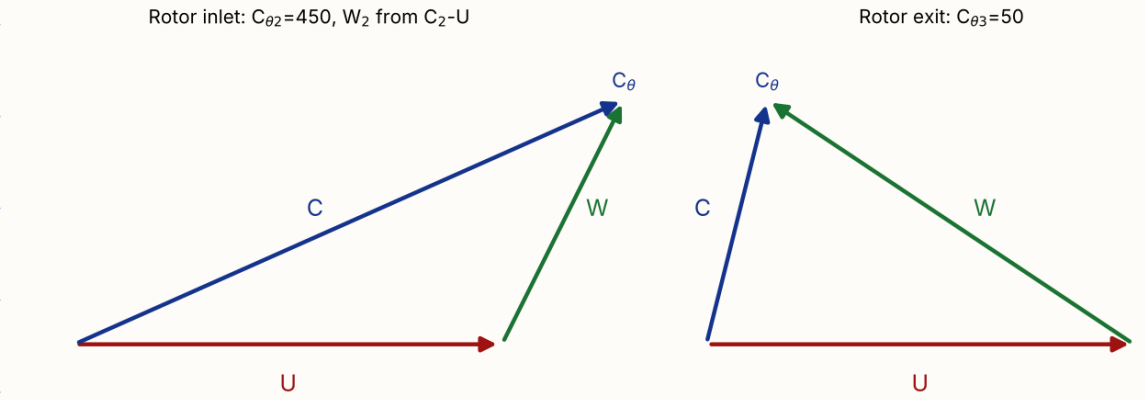
Turbo Machinery | Handwritten-style exam notes | 27 Oct 2026

Syllabus covered: Two-dimensional pitch-line design; stage loading capacity; degree of reaction; stage efficiency; turbine performance; blade cooling; examples. CFD: governing equations, numerical methods, test cases for flow and heat transfer.

A. Axial flow turbine

1. Two-dimensional pitch-line design

Turbine rotor inlet and exit triangles



The axial turbine has a stator (nozzle) row that accelerates the flow and turns it, followed by a rotor row that extracts work. The pitch-line design

uses velocity triangles at the mean radius, with C_x constant through the stage.

Stage work $\Delta h_0 = U\,(C_{\theta 2} - C_{\theta 3})$ **Stage loading $\psi = \Delta h_0 / U^2$**

Flow coefficient $\varphi = C_x / U$

Here station 2 is the nozzle exit (rotor inlet) and station 3 is the rotor exit. C_θ is measured in the direction of rotation. The sign convention must be stated clearly in the exam, because the turbine has opposite sign to the compressor.

2. Stage loading capacity

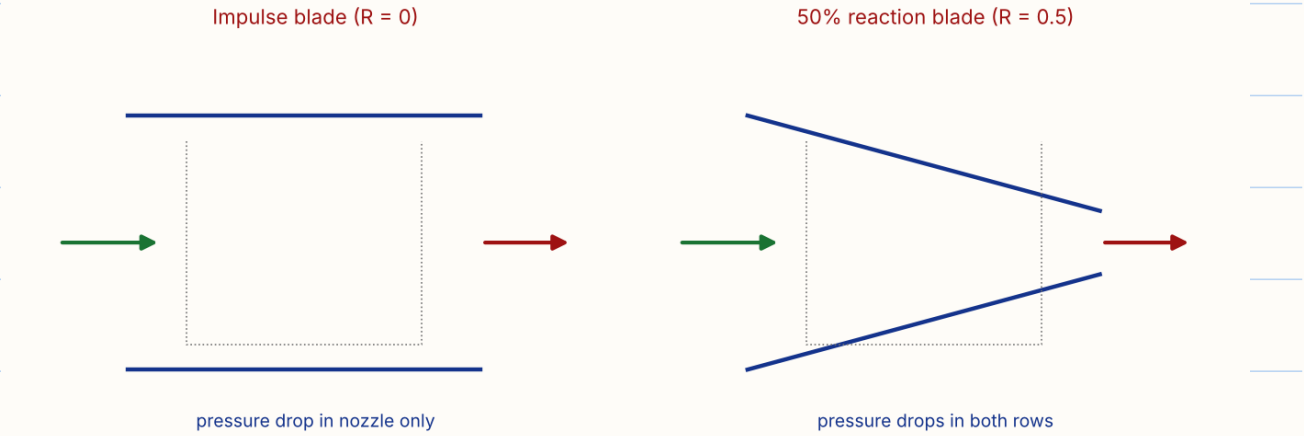
Stage loading capacity is how much work one stage can extract for a given blade speed. For high loading, the flow must turn a lot. Turbine stages can be loaded more heavily than compressor stages because the flow accelerates and separation is less of a risk. Typical ψ for a turbine stage is about 1.0 to 2.5 (higher than for compressors).

Example: *A turbine stage has U = 350 m/s, $C_{\theta 2}$ = 450 m/s and $C_{\theta 3}$ = 50 m/s. Find Δh_0 and the stage loading.*

Solution: $\Delta h_0 = 350 \times (450 - 50) = 140$ kJ/kg. $\psi = 140\,000 / 350^2 = 140\,000 / 122\,500 = 1.14$.

3. Degree of reaction and stage efficiency

Impulse and 50% reaction blades



Degree of reaction for a turbine is the fraction of the stage enthalpy drop that happens in the rotor:

$$R = \Delta h_{rotor} / \Delta h_{stage} = (W_2^2 - W_3^2) / (2\,U\,(C_{\theta 2} - C_{\theta 3}))$$

- **Impulse stage** (R ≈ 0): the expansion happens in the nozzle only. The rotor just turns the flow.
- **50% reaction stage** (R = 0.5): equal expansion in nozzle and rotor, symmetric blades, widely used.
- Higher reaction gives more expansion in rotor, which needs more blade tip clearance and gives higher stage loading in the rotor.

Stage efficiency compares the actual stage work with the isentropic work:

$\eta_{stage} = \Delta h_0 / \Delta h_{0s}$ (total-to-total or total-to-static, state which one)

4. Turbine performance

Turbine performance is shown on a characteristic map. The key variables are corrected mass flow, pressure ratio, speed and efficiency. Important features:

- The nozzle chokes when the pressure ratio is high enough for sonic flow at the throat. Beyond this point mass flow is fixed by the nozzle throat area.
- Turbine mass flow depends on inlet pressure and temperature. Corrected mass flow is used to compare operation at different inlet conditions.
- Off-design: at lower speed or pressure ratio, incidence changes and the efficiency falls.
- Losses: profile, secondary flow, tip leakage (in the rotor), trailing edge loss and cooling-air mixing losses.

Parameter	Effect on turbine	Remark

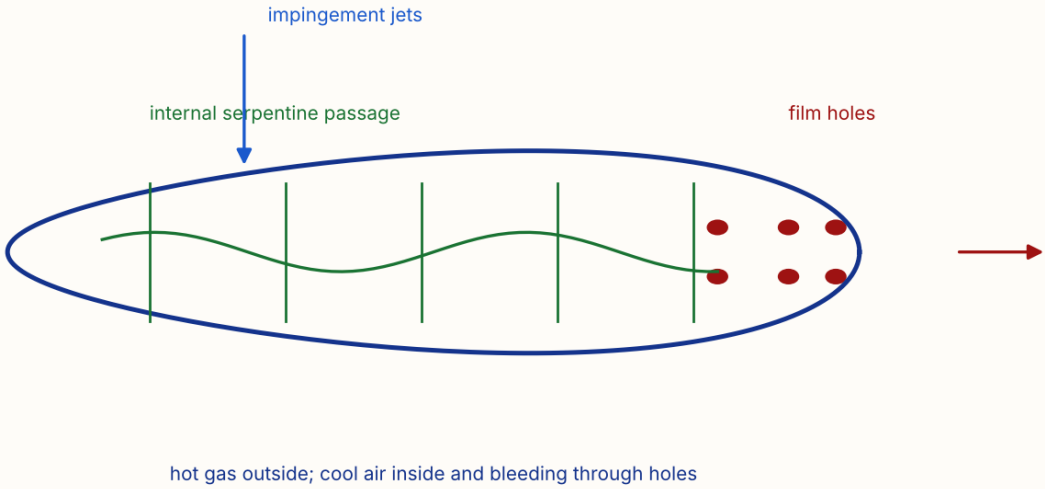
Higher pressure ratio	More work, nozzle may choke	Limit set by choking
Higher speed	Higher U, changes incidence	Efficiency peaks near design speed
Cooling air bleed	Less mass flow through rotor	Reduces cycle efficiency

5. Why blade cooling is needed

Turbine inlet temperature in modern engines is 1600 to 1800 K. Blade metal can handle only about 1100 to 1200 K. Cooling is essential to keep the blades alive. Higher inlet temperature increases specific work and cycle efficiency, so cooling technology allows a higher temperature to be used.

6. Blade cooling methods

Blade cooling methods



- **Internal convection cooling:** air flows through serpentine passages inside the blade and removes heat from the inner surface.
- **Impingement cooling:** jets of cool air strike the inner wall of the blade leading edge at high heat transfer rates.
- **Film cooling:** cool air exits through holes on the blade surface and forms a protective layer over the outer surface.
- **Transpiration cooling:** air passes through porous walls. Gives uniform film cover but is hard to manufacture.
- **Thermal barrier coating (TBC):** ceramic layer on the blade surface that insulates the metal.

Film cooling effectiveness $\eta_{fc} = (T_g - T_{aw}) / (T_g - T_c)$

Example: *Hot gas is at T_g = 1700 K, coolant at T_c = 800 K and effectiveness η_{fc} = 0.5. Find the adiabatic wall temperature T_{aw}.*

Solution: $T_{aw} = T_g - \eta_{fc}(T_g - T_c) = 1700 - 0.5 \times 900 = 1250$ K. This is above the metal limit, so more cooling or a TBC is needed.

Cost of cooling: cooling air is taken from the compressor and does not produce work in the turbine, so it reduces cycle efficiency. Designers balance higher inlet temperature against cooling air requirement.

B. CFD applied to turbomachinery

7. Governing equations

CFD solves the conservation equations for mass, momentum and energy. For a compressible viscous flow in an inertial (stationary) frame, in conservative form:

Continuity: $\partial\rho/\partial t + \nabla\cdot(\rho \mathbf{V}) = 0$ **Momentum:** $\partial(\rho\mathbf{V})/\partial t + \nabla\cdot(\rho\mathbf{V}\mathbf{V}) = -\nabla p + \nabla\cdot\tau + \mathbf{S_M}$ **Energy:** $\partial(\rho E)/\partial t + \nabla\cdot(\mathbf{V}(\rho E + p)) = \nabla\cdot(k\nabla T) + \nabla\cdot(\tau\cdot\mathbf{V})$

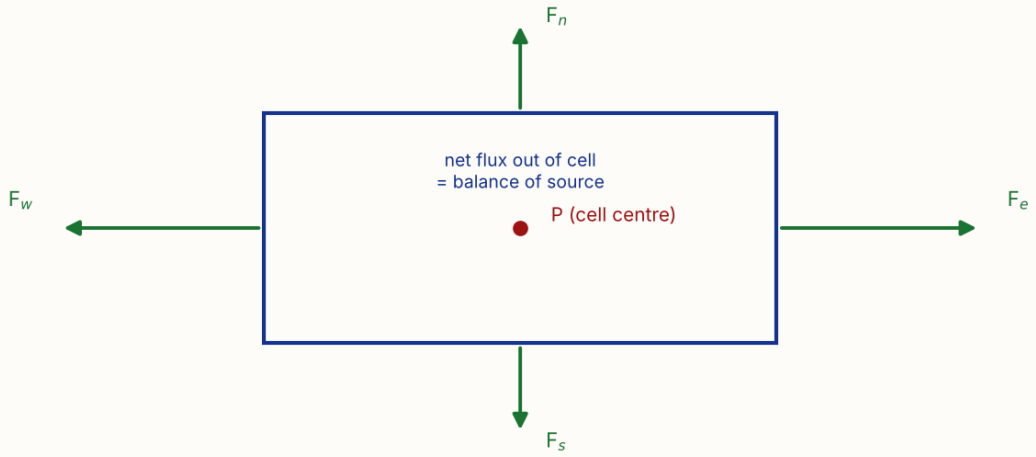
In a rotating blade row it is common to write the equations in the rotating frame. Extra source terms appear: the Coriolis term $-2\rho(\Omega \times W)$ and the

centrifugal term $-\rho\Omega \times (\Omega \times r)$. These are the same terms derived in Unit 1.

Turbulence: real turbomachinery flows are turbulent. Reynolds-averaged Navier-Stokes (RANS) equations are used, with a turbulence model. Common models are the k-ε model (good in free-stream) and the k-ω SST model (better near walls and for separation). LES and DNS are used for research because they are expensive.

8. Numerical methods

Finite volume cell



- **Finite volume method:** most common in turbomachinery CFD. The domain is divided into control volumes, and conservation is enforced on each one.
- **Finite difference method:** used for structured grids, simple geometry and research codes.

- **Finite element method:** more common in structural analysis, sometimes used for heat transfer.
- **Discretisation schemes:** upwind (stable, diffusive) and central differences (accurate, can oscillate). Second-order schemes are standard for final results.

Time marching: steady-state problems are solved by iterating in pseudo-time until residuals drop. Unsteady problems are solved with time-accurate methods. Explicit schemes need a small time step, controlled by the CFL number:

CFL = u Δt / Δx ≤ 1 (for explicit schemes, stability limit)

Example: *Grid spacing Δx = 0.1 mm and velocity u = 200 m/s. Find the maximum time step for CFL = 1.*

Solution: $\Delta t \leq \Delta x / u = 0.1 \times 10^{-3} / 200 = 0.5$ μs.

9. Grids and modelling of rotating blades

Structured grids (H-grid, O-grid, C-grid) are popular for blades because they give good control of wall resolution. **Unstructured grids** are flexible for complex geometry. Near-wall resolution is measured by y+:

$y+ = u_\tau\, y / \nu$ ($y+ \approx 1$ for wall-resolved; $y+ \approx 30$ to 100 for wall-function models)

Stage interaction between rotor and stator is handled by a **mixing plane** (circumferential average at the interface) or a **sliding mesh** (rotor moves relative to stator, unsteady). Frozen rotor is a steady approximation that fixes the relative position.

10. Test cases

Test cases are used to check the solver and the model before using it for design. Good test cases for turbomachinery flow and heat transfer are:

- **Flat plate boundary layer:** checks skin friction and heat transfer (Stanton number) against correlations.
- **Turbine or compressor cascade:** checks pressure distribution, loss coefficient and flow angles against measured data.
- **Single stage or multi-stage test rig:** checks total pressure ratio, efficiency and mass flow over a range of speeds.
- **Blade heat transfer:** checks heat transfer coefficient on the blade surface, including film-cooled cases.

Results are validated by comparing with experimental data and by grid-independence studies (refining the grid until the solution no longer changes).

11. Exam quick points

- Turbine: $\Delta h_0 = U(C_{\theta 2} - C_{\theta 3})$. Stage loading ψ can be 1 to 2.5.
- Reaction $R = \Delta h_{rotor} / \Delta h_{stage}$. Impulse $R \approx 0$, 50% reaction $R = 0.5$.
- Blade cooling: internal convection, impingement, film cooling, TBC. $\eta_{fc} = (T_g - T_{aw}) / (T_g - T_c)$.
- CFD: continuity, momentum, energy. RANS with k-ω SST. Finite volume most common. CFL ≤ 1 for explicit. y+ controls near-wall grid.

12. Practice questions

Q1. Draw a velocity triangle for a 50% reaction turbine stage and explain why the stator and rotor have symmetric blades.

Q2. Explain the difference between film cooling and impingement cooling, and when each is used.

Q3. A turbine stage has $\Delta h_0 = 120$ kJ/kg and U = 300 m/s. Find the stage loading.

Q4. State the three conservation equations solved in CFD and explain why RANS is used for turbomachinery design.